

Generative AI Engineering

Course ID	COURSE-029778	Duration	TBD
Delivery Mode	TBD	Platform	LMS package (SCORM/HTML5)
Audience / Prereqs	Software Engineers	Assessment Policy	N/A

Course Learning Outcomes (CLOs)

CLO ID	Theme	Behavior / Title	Performance Criterion	Min
CLO-01	AI Model Engineering In St Marys	Analyze AI model architectures	Meets Proficient level (3/4) on analytical rubric	60
CLO-02	AI System Design	Create scalable AI systems	Meets Proficient level (3/4) on design/production rubric	80
CLO-03	AI Application Deployment	Evaluate deployment strategies	Meets Proficient level (3/4) on analytical rubric	60
CLO-04	AI Product Optimization	Analyze AI product performance	Meets Proficient level (3/4) on analytical rubric	60
CLO-05	AI Monetization Strategies	Create monetization models	Meets Proficient level (3/4) on design/production rubric	60

Module Learning Outcomes (MLOs)

MLO ID	Behavior / Title	Performance Criterion	CLO Crosswalk	Min
M1-LO1	Analyze the architecture of transformer models New Test	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M1-LO2	Evaluate the effectiveness of attention mechanisms in LLMs	Meets Proficient level (3/4) on analytical rubric	CLO-01	60
M1-LO3	Apply tokenization techniques to preprocess text data	≥75% on structured application exercise	CLO-01	40
M2-LO1	Analyze the architecture of OpenAI's API	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M2-LO2	Create a scalable AI system using OpenAI's API	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M2-LO3	Evaluate deployment strategies for AI models using Claude	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M3-LO1	Analyze the effectiveness of zero-shot prompting techniques	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M3-LO2	Evaluate the performance of few-shot prompting in AI models	Meets Proficient level (3/4) on analytical rubric	CLO-01, CLO-04	60
M3-LO3	Create prompt chains to enhance AI model responses	Meets Proficient level (3/4) on design/production rubric	CLO-02	75
M4-LO1	Analyze the architecture of a streaming ChatGPT clone	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M4-LO2	Create a scalable session management system for a ChatGPT clone	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M4-LO3	Evaluate the integration of frontend and backend components in a streaming ChatGPT clone	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M5-LO1	Analyze AI application architecture components	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M5-LO2	Create a scalable AI system design	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M5-LO3	Evaluate AI system performance using benchmarking metrics	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M6-LO1	Analyze the components of RAG systems	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M6-LO2	Create a document processing workflow for RAG systems	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M6-LO3	Evaluate the quality of embeddings in RAG systems	Meets Proficient level (3/4) on analytical rubric	CLO-04	50
M7-LO1	Analyze the architecture of vector databases	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M7-LO2	Implement similarity search using vector databases	≥75% on structured application exercise	CLO-02	60

MLO ID	Behavior / Title	Performance Criterion	CLO Crosswalk	Min
M7-LO3	Evaluate the effectiveness of metadata filtering in vector databases	Meets Proficient level (3/4) on analytical rubric	CLO-04	50
M8-LO1	Design a queue system for ingestion pipelines	Meets Proficient level (3/4) on design/production rubric	CLO-02	120
M8-LO2	Evaluate the reliability of ingestion pipelines	Meets Proficient level (3/4) on analytical rubric	CLO-03	90
M8-LO3	Optimize ingestion pipeline scalability	Meets Proficient level (3/4) on design/production rubric	CLO-02, CLO-04	150
M9-LO1	Analyze chunk boundary issues in vectorless retrieval	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M9-LO2	Evaluate alternatives to vector retrieval methods	Meets Proficient level (3/4) on analytical rubric	CLO-02	60
M9-LO3	Analyze the effectiveness of graph search in vectorless retrieval	Meets Proficient level (3/4) on analytical rubric	CLO-01, CLO-04	50
M10-LO1	Analyze the differences between chain-based and agent-based models	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M10-LO2	Create a basic agent planning algorithm	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M10-LO3	Evaluate the execution efficiency of agent-based systems	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M11-LO1	Analyze the architecture of LangGraph within state machines	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M11-LO2	Design agent workflows using LangGraph	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M11-LO3	Evaluate durable execution strategies in LangGraph	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M12-LO1	Analyze the architecture of OpenAI Agents SDK	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M12-LO2	Create a scalable AI system using the OpenAI Agents SDK	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M12-LO3	Evaluate deployment strategies for AI applications using the OpenAI Agents SDK	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M13-LO1	Analyze the architecture of the Claude ecosystem in Nagaed	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M13-LO2	Create a prompt caching strategy within the Claude ecosystem	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M13-LO3	Evaluate the effectiveness of extended thinking in the Claude ecosystem	Meets Proficient level (3/4) on analytical rubric	CLO-04	50
M14-LO1	Analyze the architecture of voice AI systems	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M14-LO2	Create a scalable voice AI system for real-time interactions	Meets Proficient level (3/4) on design/production rubric	CLO-02	60

MLO ID	Behavior / Title	Performance Criterion	CLO Crosswalk	Min
M14-LO3	Evaluate deployment strategies for voice AI applications	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M15-LO1	Analyze the architecture of memory systems	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M15-LO2	Create a scalable implementation of a knowledge graph	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M15-LO3	Evaluate the performance of memory systems	Meets Proficient level (3/4) on analytical rubric	CLO-04	50
M16-LO1	Analyze the architecture of the Model Context Protocol	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M16-LO2	Create a scalable client-server model using MCP	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M16-LO3	Evaluate the deployment strategies of MCP servers	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M17-LO1	Analyze the impact of observability on AI model performance	Meets Proficient level (3/4) on analytical rubric	CLO-01	45
M17-LO2	Create a cost optimization plan for AI systems	Meets Proficient level (3/4) on design/production rubric	CLO-02	60
M17-LO3	Evaluate security measures in AI product deployment	Meets Proficient level (3/4) on analytical rubric	CLO-03	50
M18-LO1	Analyze monetization strategies for SaaS products	Meets Proficient level (3/4) on analytical rubric	CLO-05	45
M18-LO2	Evaluate payment system integration for AI products	Meets Proficient level (3/4) on analytical rubric	CLO-05	40
M18-LO3	Create a credit system model for AI monetization	Meets Proficient level (3/4) on design/production rubric	CLO-05	60

Module & Unit Structure

Level	Code	Title	Description	Time	Assessment
Module	M1	Phase 1 - LLM Foundations	4 units	120 min	Comparative Analysis
<i>Unit</i>	<i>1.1</i>	<i>Analyzing Transformer Model Architecture</i>	This unit covers the detailed architecture of transformer models, including their components and functionalities.	<i>40 min</i>	<i>Quiz</i>
Subtopic	1.1.1	Introduction to Transformers	Overview of transformer models and their significance in LLMs.	15 min	
Subtopic	1.1.2	Components of Transformer Models	Detailed exploration of the key components of transformer models.	15 min	
Subtopic	1.1-S3	Aonok Test	Test	60 min	
Subtopic	1.1-S4	New Test	Test	60 min	
<i>Unit</i>	<i>1.2</i>	<i>Evaluating Attention Mechanisms</i>	This unit focuses on evaluating the effectiveness of attention mechanisms within LLMs.	<i>40 min</i>	<i>Reflection</i>
Subtopic	1.2.1	Understanding Attention Mechanisms	Exploration of how attention mechanisms function within transformer models.	15 min	
Subtopic	1.2.2	Effectiveness of Attention Mechanisms	Evaluation of the impact and effectiveness of attention mechanisms in LLMs.	15 min	
<i>Unit</i>	<i>1.3</i>	<i>Applying Tokenization Techniques</i>	This unit covers the application of various tokenization techniques for text preprocessing.	<i>40 min</i>	<i>Practice</i>
Subtopic	1.3.1	Introduction to Tokenization	Basics of tokenization and its role in text preprocessing.	15 min	
Subtopic	1.3.2	Techniques for Tokenization	Various techniques used for tokenizing text data.	15 min	
<i>Unit</i>	<i>M1-U4</i>	<i>NEW</i>	<i>NEWWWW</i>	<i>60 min</i>	<i>Assignment</i>
Module	M2	Phase 2 - AI APIs & Models	3 units	120 min	Design Artifact
<i>Unit</i>	<i>2.1</i>	<i>Analyzing OpenAI API Architecture</i>	This unit covers the analysis of OpenAI's API architecture, focusing on its components and functionalities.	<i>40 min</i>	<i>Quiz</i>
Subtopic	2.1.1	API Components	Understanding the key components of OpenAI's API.	15 min	
Subtopic	2.1.2	API Functionalities	Exploring the functionalities provided by OpenAI's API.	15 min	
<i>Unit</i>	<i>2.2</i>	<i>Creating Scalable AI Systems</i>	This unit focuses on creating scalable AI systems using OpenAI's API.	<i>40 min</i>	<i>Practice Exercise</i>
Subtopic	2.2.1	System Design Principles	Principles for designing scalable AI systems.	20 min	
Subtopic	2.2.2	Integration with OpenAI API	Integrating systems with OpenAI's API for scalability.	20 min	
<i>Unit</i>	<i>2.3</i>	<i>Evaluating AI Model Deployment Strategies</i>	This unit evaluates various deployment strategies for AI models using Claude.	<i>40 min</i>	<i>Reflection Prompt</i>
Subtopic	2.3.1	Deployment Strategies Overview	Overview of different deployment strategies for AI models.	20 min	

Level	Code	Title	Description	Time	Assessment
Subtopic	2.3.2	Evaluating Strategy Effectiveness	Evaluating the effectiveness of deployment strategies.	20 min	
Module	M3	Phase 3 - Prompt Engineering	3 units	120 min	Design Artifact
<i>Unit</i>	3.1	<i>Analyzing Zero-shot Prompting Techniques</i>	This unit covers the analysis of zero-shot prompting techniques and their effectiveness in AI models.	40 min	Quiz
Subtopic	3.1.1	Introduction to Zero-shot Prompting	Understanding the basics and applications of zero-shot prompting.	15 min	
Subtopic	3.1.2	Effectiveness of Zero-shot Techniques	Analyzing the effectiveness of zero-shot techniques in various scenarios.	15 min	
<i>Unit</i>	3.2	<i>Evaluating Few-shot Prompting Performance</i>	This unit focuses on evaluating the performance of few-shot prompting in AI models.	40 min	Reflection
Subtopic	3.2.1	Understanding Few-shot Prompting	Exploring the concept and use cases of few-shot prompting.	15 min	
Subtopic	3.2.2	Performance Metrics for Few-shot Techniques	Evaluating the performance metrics used to assess few-shot techniques.	15 min	
<i>Unit</i>	3.3	<i>Creating Effective Prompt Chains</i>	This unit involves creating prompt chains to enhance AI model responses.	40 min	Practice Exercise
Subtopic	3.3.1	Basics of Prompt Chaining	Introduction to the concept and benefits of prompt chaining.	15 min	
Subtopic	3.3.2	Designing Effective Prompt Chains	Techniques and strategies for designing effective prompt chains.	15 min	
Module	M4	Phase 4 - Streaming ChatGPT Clone	3 units	120 min	Recommendation Memo
<i>Unit</i>	4.1	<i>Analyzing the Architecture of a Streaming ChatGPT Clone</i>	This unit covers the architectural components and design patterns used in a streaming ChatGPT clone.	40 min	Quiz
Subtopic	4.1.1	Streaming UI Architecture	Exploration of the UI components and their interactions in a streaming environment.	15 min	
Subtopic	4.1.2	Backend Architecture	Understanding the backend services and their roles in streaming data.	15 min	
<i>Unit</i>	4.2	<i>Creating a Scalable Session Management System</i>	This unit focuses on designing and implementing a scalable session management system for a ChatGPT clone.	40 min	Practice Exercise
Subtopic	4.2.1	Session State Management	Techniques for maintaining session state across distributed systems.	20 min	
Subtopic	4.2.2	Scalability Considerations	Strategies for ensuring scalability in session management.	20 min	
<i>Unit</i>	4.3	<i>Evaluating Integration of Frontend and Backend Components</i>	This unit examines the integration points between frontend and backend components in a streaming ChatGPT clone.	40 min	Reflection Prompt
Subtopic	4.3.1	Frontend-Backend Communication	Methods and protocols for effective communication between frontend and backend.	20 min	

Level	Code	Title	Description	Time	Assessment
Subtopic	4.3.2	Integration Testing	Approaches to testing the integration of system components.	20 min	
Module	M5	AI System Design	3 units	120 min	Design Artifact
<i>Unit</i>	5.1	<i>Analyzing AI Application Architecture</i>	This unit covers the analysis of AI application architecture components.	40 min	Quiz
Subtopic	5.1.1	Introduction to AI Architecture	Overview of AI application architecture and its importance.	15 min	
Subtopic	5.1.2	Components of AI Systems	Detailed analysis of different components within AI systems.	15 min	
<i>Unit</i>	5.2	<i>Creating Scalable AI System Designs</i>	This unit focuses on creating designs for scalable AI systems.	40 min	Practice Exercise
Subtopic	5.2.1	Principles of Scalability	Understanding the principles that ensure scalability in AI systems.	20 min	
Subtopic	5.2.2	Designing for Scalability	Techniques and strategies for designing scalable AI systems.	20 min	
<i>Unit</i>	5.3	<i>Evaluating AI System Performance</i>	This unit covers the evaluation of AI system performance using benchmarking metrics.	40 min	Reflection Prompt
Subtopic	5.3.1	Benchmarking Metrics	Introduction to various benchmarking metrics used in AI.	20 min	
Subtopic	5.3.2	Performance Evaluation Techniques	Methods to evaluate and improve AI system performance.	20 min	
Module	M6	Phase 6 - RAG Systems	3 units	120 min	Design Artifact
<i>Unit</i>	6.1	<i>Analyzing Components of RAG Systems</i>	This unit covers the analysis of various components that make up RAG systems.	40 min	Quiz
Subtopic	6.1.1	Introduction to RAG Systems	Overview of RAG systems and their importance.	15 min	
Subtopic	6.1.2	Key Components of RAG Systems	Detailed analysis of the components involved in RAG systems.	15 min	
<i>Unit</i>	6.2	<i>Creating Document Processing Workflows</i>	This unit focuses on designing workflows for document processing within RAG systems.	40 min	Practice Exercise
Subtopic	6.2.1	Document Processing Techniques	Exploration of various techniques used in document processing.	20 min	
Subtopic	6.2.2	Workflow Design Principles	Principles and best practices for designing effective workflows.	20 min	
<i>Unit</i>	6.3	<i>Evaluating Embedding Quality</i>	This unit addresses the evaluation of embedding quality in RAG systems.	40 min	Reflection
Subtopic	6.3.1	Understanding Embeddings	Introduction to embeddings and their role in RAG systems.	20 min	
Subtopic	6.3.2	Quality Metrics for Embeddings	Evaluation metrics used to assess embedding quality.	20 min	
Module	M7	Phase 7 - Vector Databases	3 units	120 min	Recommendation Memo

Level	Code	Title	Description	Time	Assessment
Unit	7.1	<i>Analyzing Vector Database Architecture</i>	This unit covers the fundamental architecture of vector databases and their components.	40 min	Quiz
Subtopic	7.1.1	Introduction to Vector Databases	Overview of vector databases and their role in data management.	15 min	
Subtopic	7.1.2	Components of Vector Databases	Detailed look at the components that make up vector databases.	15 min	
Unit	7.2	<i>Implementing Similarity Search</i>	This unit focuses on the implementation of similarity search using vector databases.	40 min	Practice Exercise
Subtopic	7.2.1	Techniques for Similarity Search	Exploration of various techniques used for similarity search in vector databases.	20 min	
Subtopic	7.2.2	Hands-on Implementation	Practical implementation of similarity search in a vector database.	20 min	
Unit	7.3	<i>Evaluating Metadata Filtering</i>	This unit evaluates the effectiveness of metadata filtering in vector databases.	40 min	Reflection Prompt
Subtopic	7.3.1	Understanding Metadata Filtering	Introduction to metadata filtering and its applications.	20 min	
Subtopic	7.3.2	Evaluating Filtering Techniques	Assessment of different metadata filtering techniques.	20 min	
Module	M8	Phase 8 - Production Ingestion Pipelines	3 units	120 min	Design Artifact
Unit	8.1	<i>Designing Queue Systems for Ingestion Pipelines</i>	This unit covers the principles and practices of designing queue systems tailored for ingestion pipelines.	40 min	Practice exercise
Subtopic	8.1.1	Introduction to Queue Systems	Overview of queue systems and their role in ingestion pipelines.	15 min	
Subtopic	8.1.2	Design Principles for Queue Systems	Explore design principles and best practices for queue systems.	15 min	
Unit	8.2	<i>Evaluating Reliability of Ingestion Pipelines</i>	This unit focuses on assessing the reliability of ingestion pipelines and identifying potential improvements.	40 min	Reflection prompt
Subtopic	8.2.1	Reliability Metrics	Learn about key metrics used to evaluate pipeline reliability.	20 min	
Subtopic	8.2.2	Improving Pipeline Reliability	Strategies and techniques to enhance the reliability of ingestion pipelines.	20 min	
Unit	8.3	<i>Optimizing Ingestion Pipeline Scalability</i>	This unit explores methods to optimize the scalability of ingestion pipelines.	40 min	Peer discussion
Subtopic	8.3.1	Scalability Challenges	Identify common challenges in scaling ingestion pipelines.	20 min	
Subtopic	8.3.2	Techniques for Scalability	Explore techniques to enhance pipeline scalability.	20 min	
Module	M9	Phase 9 - Vectorless Retrieval	3 units	120 min	Comparative Analysis
Unit	9.1	<i>Analyzing Chunk Boundary Issues</i>	This unit covers the analysis of chunk boundary issues in vectorless retrieval.	40 min	Quiz

Level	Code	Title	Description	Time	Assessment
Subtopic	9.1.1	Understanding Chunk Boundaries	Explore the concept of chunk boundaries in data retrieval.	15 min	
Subtopic	9.1.2	Challenges in Chunk Boundary Identification	Analyze the challenges faced in identifying chunk boundaries.	15 min	
Unit	9.2	<i>Evaluating Alternatives to Vector Retrieval</i>	This unit evaluates various alternatives to traditional vector retrieval methods.	40 min	Reflection
Subtopic	9.2.1	Overview of Alternative Methods	Examine different methods that serve as alternatives to vector retrieval.	20 min	
Subtopic	9.2.2	Pros and Cons of Alternatives	Evaluate the advantages and disadvantages of alternative retrieval methods.	20 min	
Unit	9.3	<i>Analyzing Graph Search Effectiveness</i>	This unit analyzes the effectiveness of graph search in vectorless retrieval.	40 min	Practice Exercise
Subtopic	9.3.1	Introduction to Graph Search	Learn about graph search techniques used in retrieval.	20 min	
Subtopic	9.3.2	Effectiveness of Graph Search	Analyze how effective graph search is in vectorless retrieval.	20 min	
Module	M10	Agent Fundamentals	3 units	120 min	Design Artifact
Unit	10.1	<i>Analyzing Chain vs Agent Models</i>	This unit covers the analysis of differences between chain-based and agent-based models.	40 min	Quiz
Subtopic	10.1.1	Introduction to Chain-Based Models	Explore the fundamentals of chain-based models.	15 min	
Subtopic	10.1.2	Introduction to Agent-Based Models	Explore the fundamentals of agent-based models.	15 min	
Subtopic	10.1.3	Comparative Analysis	Analyze the differences between chain-based and agent-based models.	10 min	
Unit	10.2	<i>Creating Agent Planning Algorithms</i>	This unit focuses on creating basic agent planning algorithms.	40 min	Practice Exercise
Subtopic	10.2.1	Basics of Agent Planning	Understand the fundamentals of planning in agent systems.	20 min	
Subtopic	10.2.2	Algorithm Development	Learn to create basic planning algorithms for agents.	20 min	
Unit	10.3	<i>Evaluating Execution Efficiency</i>	This unit evaluates the execution efficiency of agent-based systems.	40 min	Reflection Prompt
Subtopic	10.3.1	Understanding Execution Efficiency	Learn what execution efficiency means in the context of agent systems.	15 min	
Subtopic	10.3.2	Measuring Efficiency	Explore methods to measure execution efficiency in agent systems.	15 min	
Subtopic	10.3.3	Improving Efficiency	Learn strategies to enhance execution efficiency.	10 min	
Module	M11	Phase 11 - LangGraph	3 units	120 min	Recommendation Memo
Unit	11.1	<i>Analyzing LangGraph Architecture in State Machines</i>	This unit covers the analysis of LangGraph's architecture within the context of state machines.	40 min	Quiz

Level	Code	Title	Description	Time	Assessment
Subtopic	11.1.1	Introduction to State Machines	Overview of state machines and their role in LangGraph.	15 min	
Subtopic	11.1.2	LangGraph Architecture	Detailed analysis of LangGraph's architecture.	15 min	
Subtopic	11.1.3	Case Studies	Analysis of real-world applications of LangGraph in state machines.	10 min	
Unit	11.2	<i>Designing Agent Workflows with LangGraph</i>	This unit focuses on designing effective agent workflows using LangGraph.	40 min	Practice Exercise
Subtopic	11.2.1	Basics of Agent Workflows	Understanding the fundamentals of agent workflows.	15 min	
Subtopic	11.2.2	LangGraph Workflow Tools	Exploration of tools and features in LangGraph for workflow design.	15 min	
Subtopic	11.2.3	Advanced Workflow Design	Techniques for designing complex workflows using LangGraph.	10 min	
Unit	11.3	<i>Evaluating Durable Execution Strategies in LangGraph</i>	This unit evaluates strategies for durable execution within LangGraph.	40 min	Reflection Prompt
Subtopic	11.3.1	Understanding Durable Execution	Introduction to durable execution and its importance.	15 min	
Subtopic	11.3.2	Strategies for Durable Execution	Evaluation of different strategies for achieving durable execution.	15 min	
Subtopic	11.3.3	Implementation Challenges	Challenges faced during the implementation of durable execution strategies.	10 min	
Module	M12	OpenAI Agents SDK	3 units	120 min	Design Artifact
Unit	12.1	<i>Analyzing the Architecture of OpenAI Agents SDK</i>	This unit covers the detailed architecture of the OpenAI Agents SDK, focusing on its components and their interactions.	40 min	Quiz
Subtopic	12.1.1	SDK Components	Explore the core components of the OpenAI Agents SDK.	15 min	
Subtopic	12.1.2	Component Interactions	Analyze how different components interact within the SDK.	15 min	
Unit	12.2	<i>Creating a Scalable AI System</i>	This unit focuses on designing and creating a scalable AI system using the OpenAI Agents SDK.	40 min	Practice Exercise
Subtopic	12.2.1	Design Principles	Learn the principles of designing scalable AI systems.	20 min	
Subtopic	12.2.2	Implementation Strategies	Explore strategies for implementing scalable systems.	20 min	
Unit	12.3	<i>Evaluating Deployment Strategies</i>	This unit covers the evaluation of various deployment strategies for AI applications using the OpenAI Agents SDK.	40 min	Reflection
Subtopic	12.3.1	Deployment Options	Review different deployment options available for AI applications.	20 min	
Subtopic	12.3.2	Performance Evaluation	Evaluate the performance of deployed AI systems.	20 min	
Module	M13	Claude Ecosystem	3 units	120 min	Design Artifact

Level	Code	Title	Description	Time	Assessment
Unit	13.1	<i>Analyzing the Claude Ecosystem Architecture</i>	This unit covers the detailed analysis of the Claude ecosystem's architecture.	40 min	Quiz
Subtopic	13.1.1	Claude System Components	Overview of the main components of the Claude ecosystem.	15 min	
Subtopic	13.1.2	Integration and Interactions	How different components of the Claude ecosystem interact and integrate.	15 min	
Unit	13.2	<i>Creating a Prompt Caching Strategy</i>	This unit focuses on developing a strategy for prompt caching within the Claude ecosystem.	40 min	Practice Exercise
Subtopic	13.2.1	Understanding Prompt Caching	Basics of prompt caching and its importance.	15 min	
Subtopic	13.2.2	Designing a Caching Strategy	Steps to design an effective prompt caching strategy.	15 min	
Unit	13.3	<i>Evaluating Extended Thinking Effectiveness</i>	This unit evaluates the effectiveness of extended thinking within the Claude ecosystem.	40 min	Reflection Prompt
Subtopic	13.3.1	Concept of Extended Thinking	Introduction to extended thinking and its role in the ecosystem.	15 min	
Subtopic	13.3.2	Measuring Effectiveness	Methods to evaluate the effectiveness of extended thinking.	15 min	
Module	M14	Voice AI	3 units	120 min	Design Artifact
Unit	14.1	<i>Analyzing Voice AI System Architecture</i>	This unit covers the analysis of the architecture of voice AI systems.	40 min	Quiz
Subtopic	14.1.1	Components of Voice AI Systems	Explore the key components that make up voice AI systems.	15 min	
Subtopic	14.1.2	System Architecture Patterns	Learn about different architecture patterns used in voice AI.	15 min	
Unit	14.2	<i>Creating Scalable Voice AI Systems</i>	This unit focuses on creating scalable voice AI systems for real-time interactions.	40 min	Practice Exercise
Subtopic	14.2.1	Scalability Challenges	Understand the challenges in scaling voice AI systems.	15 min	
Subtopic	14.2.2	Real-time Interaction Design	Designing systems for effective real-time interactions.	15 min	
Unit	14.3	<i>Evaluating Deployment Strategies for Voice AI</i>	This unit evaluates different deployment strategies for voice AI applications.	40 min	Reflection
Subtopic	14.3.1	Deployment Models	Explore various deployment models for voice AI.	15 min	
Subtopic	14.3.2	Security and Compliance	Evaluate security and compliance considerations in deployment.	15 min	
Module	M15	Phase 15 - Memory Systems	3 units	120 min	Design Artifact
Unit	15.1	<i>Analyzing Memory System Architectures</i>	This unit covers the analysis of different memory system architectures, focusing on their components and interactions.	40 min	Quiz
Subtopic	15.1.1	Short-term Memory Architecture	Exploration of the components and functions of short-term memory systems.	15 min	

Level	Code	Title	Description	Time	Assessment
Subtopic	15.1.2	Long-term Memory Architecture	Understanding the structure and processes involved in long-term memory systems.	15 min	
Unit	15.2	<i>Creating Scalable Knowledge Graphs</i>	This unit focuses on the creation and implementation of scalable knowledge graphs within memory systems.	40 min	Practice Exercise
Subtopic	15.2.1	Introduction to Knowledge Graphs	Basics of knowledge graphs and their role in memory systems.	15 min	
Subtopic	15.2.2	Scalability in Knowledge Graphs	Techniques and considerations for ensuring scalability in knowledge graph implementations.	15 min	
Unit	15.3	<i>Evaluating Memory System Performance</i>	This unit involves evaluating the performance of various memory systems using specific criteria and metrics.	40 min	Reflection
Subtopic	15.3.1	Performance Metrics for Memory Systems	Identifying and understanding key performance metrics for memory systems.	15 min	
Subtopic	15.3.2	Case Studies in Memory Performance	Analyzing real-world case studies to evaluate memory system performance.	15 min	
Module	M16	Phase 16 - MCP	3 units	120 min	Design Artifact
Unit	16.1	<i>Analyzing MCP Architecture</i>	This unit covers the analysis of the Model Context Protocol architecture.	40 min	Quiz
Subtopic	16.1.1	Overview of MCP Architecture	Introduction to the components and structure of MCP.	15 min	
Subtopic	16.1.2	Detailed Analysis of MCP Components	In-depth analysis of each component within MCP.	15 min	
Unit	16.2	<i>Creating Scalable MCP Client-Server Models</i>	This unit focuses on creating scalable client-server models using MCP.	40 min	Practice Exercise
Subtopic	16.2.1	Design Principles for Scalability	Principles and practices for designing scalable systems.	20 min	
Subtopic	16.2.2	Implementing MCP in Client-Server Models	Practical implementation of MCP in client-server environments.	20 min	
Unit	16.3	<i>Evaluating MCP Server Deployment Strategies</i>	This unit evaluates various deployment strategies for MCP servers.	40 min	Reflection
Subtopic	16.3.1	Deployment Strategy Overview	Overview of common deployment strategies for MCP servers.	20 min	
Subtopic	16.3.2	Comparative Evaluation of Strategies	Comparing and evaluating different deployment strategies.	20 min	
Module	M17	AI Product Engineering	3 units	120 min	Design Artifact
Unit	17.1	<i>Analyzing Observability in AI Models</i>	This unit covers the impact of observability on AI model performance.	40 min	Quiz
Subtopic	17.1.1	Introduction to Observability	Understanding the basics of observability in AI systems.	15 min	
Subtopic	17.1.2	Tools and Techniques	Exploring tools and techniques for implementing observability.	15 min	
Subtopic	17.1.3	Impact on Performance	Analyzing how observability affects AI model performance.	10 min	

Level	Code	Title	Description	Time	Assessment
Unit	17.2	<i>Creating Cost Optimization Plans for AI</i>	This unit focuses on developing cost optimization strategies for AI systems.	40 min	Practice Exercise
Subtopic	17.2.1	Cost Drivers in AI	Identifying key cost drivers in AI systems.	15 min	
Subtopic	17.2.2	Optimization Techniques	Exploring techniques to optimize costs in AI deployments.	15 min	
Subtopic	17.2.3	Case Studies	Reviewing case studies of successful cost optimization.	10 min	
Unit	17.3	<i>Evaluating Security Measures in AI Deployment</i>	This unit evaluates the security measures necessary for AI product deployment.	40 min	Reflection Prompt
Subtopic	17.3.1	Security Risks in AI	Understanding potential security risks in AI systems.	15 min	
Subtopic	17.3.2	Implementing Security Protocols	Exploring protocols to secure AI deployments.	15 min	
Subtopic	17.3.3	Evaluating Security Effectiveness	Methods to evaluate the effectiveness of security measures.	10 min	
Module	M18	Monetization	3 units	120 min	Design Artifact
Unit	18.1	<i>Analyzing Monetization Strategies for SaaS Products</i>	This unit covers various monetization strategies applicable to SaaS products, focusing on their analysis.	40 min	Quiz
Subtopic	18.1.1	Introduction to SaaS Monetization	Overview of monetization in SaaS products.	15 min	
Subtopic	18.1.2	Common SaaS Monetization Models	Exploration of various monetization models used in SaaS.	15 min	
Subtopic	18.1.3	Case Studies in SaaS Monetization	Analysis of successful SaaS monetization strategies.	10 min	
Unit	18.2	<i>Evaluating Payment System Integration for AI Products</i>	This unit focuses on evaluating different payment system integrations suitable for AI products.	40 min	Reflection
Subtopic	18.2.1	Overview of Payment Systems	Introduction to various payment systems available for integration.	15 min	
Subtopic	18.2.2	Criteria for Payment System Selection	Factors to consider when selecting a payment system for AI products.	15 min	
Subtopic	18.2.3	Integration Challenges and Solutions	Common challenges faced during payment system integration and their solutions.	10 min	
Unit	18.3	<i>Creating a Credit System Model for AI Monetization</i>	This unit involves creating a credit system model tailored for AI monetization.	40 min	Practice Exercise
Subtopic	18.3.1	Understanding Credit Systems	Basic principles and components of credit systems.	15 min	
Subtopic	18.3.2	Designing a Credit System	Steps involved in designing a credit system for AI products.	15 min	
Subtopic	18.3.3	Testing and Refining Credit Models	Methods for testing and refining credit system models.	10 min	

Assessment Details

Assessment	Type / Title	Task Description	Passing	Mapped LOs
M1 - Phase 1 - LLM Foundations				
Summative	Comparative Analysis - Comparative Analysis of Transformer Models	Conduct a comparative analysis of two transformer model architectures. Identify key components, compare their functionalities, and evaluate their performance in different contexts. Provide a detailed report discussing the strengths and weaknesses of each model, supported by evidence from academic literature.	Pass: 70 / 100	CLO-01, M1-LO1, M1-LO2, M1-LO3
<i>Formative (1.1)</i>	<i>Quiz - Transformer Model Architecture Quiz</i>	Complete a 10-question multiple-choice quiz on the architecture of transformer models. Each question will test your understanding of different components and their functions within the model.	<i>Pass: 80 /</i>	<i>M1-LO1</i>
<i>Formative (1.2)</i>	<i>Reflection - Reflection on Attention Mechanisms</i>	Write a reflection on the effectiveness of attention mechanisms in large language models. Discuss how attention mechanisms contribute to model performance and provide examples from recent studies.	<i>Pass: /</i>	<i>M1-LO2</i>
<i>Formative (1.3)</i>	<i>Practice - Tokenization Techniques Practice</i>	Apply various tokenization techniques to preprocess a given text dataset. Use Python to implement tokenization and submit your code along with a brief explanation of the techniques used.	<i>Pass: /</i>	<i>M1-LO3</i>
<i>Formative (M1-U4)</i>	<i>Assignment - Assignment</i>		<i>Pass: 60 / 100</i>	
M2 - Phase 2 - AI APIs & Models				
Summative	Project - AI System Design Artifact	Design a scalable AI system using OpenAI's API. Your design should include a detailed architecture diagram, a description of the components and their interactions, and a justification for the chosen deployment strategy. Ensure your design addresses scalability, efficiency, and potential challenges.	Pass: 70 / 100	CLO-01, M2-LO1, M2-LO2, M2-LO3
<i>Formative (2.1)</i>	<i>Quiz - OpenAI API Architecture Quiz</i>	Complete a 10-question multiple-choice quiz on the architecture of OpenAI's API. Each question will test your understanding of the components and their functions within the API.	<i>Pass: 80 / 100</i>	<i>M2-LO1</i>
<i>Formative (2.2)</i>	<i>Practice Exercise - Scalable AI System Design Exercise</i>	Design a basic scalable AI system using OpenAI's API. Create a simple architecture diagram and a brief description of how the system can handle increased load.	<i>Pass: /</i>	<i>M2-LO2</i>
<i>Formative (2.3)</i>	<i>Reflection - AI Model Deployment Reflection</i>	Write a reflection on the deployment strategies for AI models using Claude. Discuss the strengths and weaknesses of different strategies and how they impact model performance.	<i>Pass: /</i>	<i>M2-LO3</i>
M3 - Phase 3 - Prompt Engineering				

Assessment	Type / Title	Task Description	Passing	Mapped LOs
Summative	Project - Design Artifact: Advanced Prompt Engineering	Design a comprehensive prompt engineering artifact that demonstrates the application of zero-shot, few-shot, and prompt chaining techniques. Include a detailed explanation of each technique, its implementation, and an evaluation of its effectiveness. Provide examples of AI model outputs for each technique and analyze the differences in performance. Conclude with recommendations for optimizing prompt strategies based on your findings.	Pass: 70 / 100	CLO-01, M3-LO1, M3-LO2, M3-LO3
<i>Formative (3.1)</i>	<i>Quiz - Zero-shot Prompting Techniques Quiz</i>	Complete a 10-question multiple-choice quiz assessing your understanding of zero-shot prompting techniques. Each question will test your ability to identify and analyze the effectiveness of these techniques in various scenarios.	<i>Pass: 80 /</i>	<i>M3-LO1</i>
<i>Formative (3.2)</i>	<i>Reflection - Few-shot Prompting Reflection</i>	Write a 500-word reflection on the performance of few-shot prompting in AI models. Discuss the strengths and limitations you observed and how these impact the model's output quality.	<i>Pass: /</i>	<i>M3-LO2</i>
<i>Formative (3.3)</i>	<i>Practice - Prompt Chain Creation Exercise</i>	Develop a series of prompt chains designed to enhance AI model responses. Test these chains with a provided AI model and document the outcomes. Reflect on the effectiveness of your chains and suggest improvements.	<i>Pass: /</i>	<i>M3-LO3</i>
M4 - Phase 4 - Streaming ChatGPT Clone				
Summative	Recommendation Memo - Integration Evaluation and Recommendation Memo	Compose a recommendation memo evaluating the integration of frontend and backend components in a streaming ChatGPT clone. Analyze the current integration setup, identify any potential issues, and propose actionable improvements. Your memo should include an executive summary, detailed evaluation, and specific recommendations supported by evidence from your analysis.	Pass: 70 / 100	M4-LO3
<i>Formative (4.1)</i>	<i>Quiz - Architecture Analysis Quiz</i>	Complete a 10-question multiple-choice quiz assessing your understanding of the architecture of a streaming ChatGPT clone. Each question will test your ability to identify and explain key architectural components and their functions.	<i>Pass: 80 /</i>	<i>M4-LO1</i>
<i>Formative (4.2)</i>	<i>Practice Exercise - Session Management System Design</i>	Design a scalable session management system for a ChatGPT clone. Create a diagram illustrating the system architecture and write a brief explanation of how your design ensures scalability and reliability.	<i>Pass: /</i>	<i>M4-LO2</i>
<i>Formative (4.3)</i>	<i>Reflection Prompt - Integration Reflection</i>	Write a reflective piece on your understanding of the integration between frontend and backend components in a streaming ChatGPT clone. Discuss any challenges you anticipate and how you plan to address them.	<i>Pass: /</i>	<i>M4-LO3</i>
M5 - AI System Design				

Assessment	Type / Title	Task Description	Passing	Mapped LOs
Summative	Project - AI System Design Artifact	Design a comprehensive AI system that addresses a specific problem in a chosen domain. Your design should include a detailed architecture diagram, scalability considerations, and performance evaluation metrics. Provide a rationale for each component and justify your design choices based on current best practices in AI system design.	Pass: 70 / 100	M5-LO2, M5-LO3
<i>Formative (5.1)</i>	<i>Quiz - AI Architecture Analysis Quiz</i>	Complete a 10-question multiple-choice quiz that tests your ability to identify and analyze components of AI application architectures. Each question will present a scenario requiring you to select the correct architectural component or principle.	<i>Pass: 80 /</i>	<i>M5-LO1</i>
<i>Formative (5.2)</i>	<i>Practice Exercise - Scalable AI Design Practice</i>	Develop a scalable design for a given AI system scenario. Your design should include a brief description of the architecture, scalability strategies, and potential challenges. Submit a 2-page document outlining your design approach.	<i>Pass: /</i>	<i>M5-LO2</i>
<i>Formative (5.3)</i>	<i>Reflection - AI Performance Evaluation Reflection</i>	Write a reflective piece on the process of evaluating AI system performance. Discuss the importance of benchmarking metrics and how they influence system design decisions. Reflect on a case study provided in class.	<i>Pass: /</i>	<i>M5-LO3</i>
M6 - Phase 6 - RAG Systems				
Summative	Design Artifact - RAG System Design Artifact	Design a comprehensive RAG (Retrieval-Augmented Generation) system artifact that includes a detailed analysis of its components, a document processing workflow, and an evaluation of embedding quality. Your design should demonstrate a clear understanding of RAG systems, effectively integrate document processing, and critically assess embedding quality. Include diagrams, flowcharts, and a written explanation of your design choices.	Pass: 70 / 100	CLO-01, M6-LO1, M6-LO2, M6-LO3
<i>Formative (6.1)</i>	<i>Quiz - RAG Components Analysis Quiz</i>	Complete a 10-question multiple-choice quiz that tests your ability to identify and analyze the key components of RAG systems. Each question presents a scenario or component description requiring you to select the correct analysis or component identification.	<i>Pass: 80 /</i>	<i>M6-LO1</i>
<i>Formative (6.2)</i>	<i>Practice Exercise - Document Processing Workflow Practice</i>	Develop a document processing workflow for a RAG system. Use provided templates to outline each step of the workflow, ensuring that it aligns with RAG system requirements. Submit your completed workflow for review.	<i>Pass: /</i>	<i>M6-LO2</i>
<i>Formative (6.3)</i>	<i>Reflection - Embedding Quality Evaluation Reflection</i>	Write a reflective piece on the process of evaluating embedding quality in RAG systems. Discuss the criteria used, challenges faced, and insights gained during the evaluation process.	<i>Pass: /</i>	<i>M6-LO3</i>
M7 - Phase 7 - Vector Databases				

Assessment	Type / Title	Task Description	Passing	Mapped LOs
Summative	Assignment - Recommendation Memo	Compose a detailed recommendation memo to a hypothetical company considering the adoption of vector databases. Analyze the architecture of vector databases, implement a similarity search example, and evaluate the effectiveness of metadata filtering. Provide a comprehensive recommendation based on your findings, supported by evidence and examples.	Pass: 70 / 100	CLO-01, M7-LO1, M7-LO2, M7-LO3
<i>Formative (7.1)</i>	<i>Quiz - Vector Database Architecture Quiz</i>	Complete a 10-question multiple-choice quiz assessing your understanding of vector database architecture. Questions will cover key components, functions, and benefits.	<i>Pass: 80 /</i>	<i>M7-LO1</i>
<i>Formative (7.2)</i>	<i>Practice - Similarity Search Implementation Exercise</i>	Implement a similarity search using a vector database. Use a provided dataset to demonstrate how to retrieve similar items based on vector similarity. Document your process and results.	<i>Pass: /</i>	<i>M7-LO2</i>
<i>Formative (7.3)</i>	<i>Reflection - Metadata Filtering Reflection</i>	Write a reflection on the effectiveness of metadata filtering in vector databases. Consider its impact on data retrieval accuracy and efficiency.	<i>Pass: /</i>	<i>M7-LO3</i>
M8 - Phase 8 - Production Ingestion Pipelines				
Summative	Project - Design Artifact for Ingestion Pipelines	Design a comprehensive ingestion pipeline system that includes queue systems, evaluates reliability, and optimizes scalability. Your design should include a detailed architecture diagram, a written explanation of the queue system design, an evaluation of reliability factors, and a scalability optimization plan. Ensure your design addresses potential bottlenecks and includes strategies for fault tolerance.	Pass: 70 / 100	CLO-02, M8-LO1, M8-LO2, M8-LO3
<i>Formative (8.1)</i>	<i>Practice - Queue System Design Exercise</i>	Design a queue system for a given ingestion pipeline scenario. Consider factors such as message throughput, latency, and fault tolerance. Submit a diagram and a brief explanation of your design choices.	<i>Pass: /</i>	<i>M8-LO1</i>
<i>Formative (8.2)</i>	<i>Reflection - Reliability Evaluation Reflection</i>	Reflect on the reliability of an ingestion pipeline you have studied. Consider factors such as data integrity, fault tolerance, and recovery mechanisms. Write a reflection on how these factors impact the overall reliability of the system.	<i>Pass: /</i>	<i>M8-LO2</i>
<i>Formative (8.3)</i>	<i>Discussion - Scalability Optimization Discussion</i>	Participate in a peer discussion on strategies for optimizing the scalability of ingestion pipelines. Share your ideas and provide feedback on peers' suggestions. Focus on load balancing, resource allocation, and system architecture.	<i>Pass: /</i>	<i>M8-LO3</i>
M9 - Phase 9 - Vectorless Retrieval				

Assessment	Type / Title	Task Description	Passing	Mapped LOs
Summative	Comparative Analysis - Comparative Analysis of Vectorless Retrieval Methods	Conduct a comprehensive comparative analysis of different vectorless retrieval methods. Your analysis should include a detailed examination of chunk boundary issues, evaluation of alternative retrieval methods, and an analysis of graph search effectiveness. Provide a critical comparison of these methods, highlighting strengths, weaknesses, and potential applications. Use case studies or examples to support your analysis.	Pass: 70 / 100	CLO-01, M9-LO1, M9-LO2, M9-LO3
<i>Formative (9.1)</i>	<i>Quiz - Chunk Boundary Issues Quiz</i>	Complete a 10-question multiple-choice quiz focused on identifying and analyzing chunk boundary issues in vectorless retrieval systems. Each question will present a scenario requiring you to determine the boundary issue and suggest potential solutions.	<i>Pass: 80 /</i>	<i>M9-LO1</i>
<i>Formative (9.2)</i>	<i>Reflection - Reflection on Alternatives to Vector Retrieval</i>	Write a reflective piece on the various alternatives to vector retrieval methods. Discuss the potential benefits and drawbacks of each method and reflect on their applicability in different contexts.	<i>Pass: /</i>	<i>M9-LO2</i>
<i>Formative (9.3)</i>	<i>Practice Exercise - Graph Search Effectiveness Analysis</i>	Engage in a practice exercise where you analyze the effectiveness of graph search techniques in vectorless retrieval. Use provided datasets to perform a basic analysis and report your findings.	<i>Pass: /</i>	<i>M9-LO3</i>
M10 - Agent Fundamentals				
Summative	Project - Agent System Design Artifact	Design a comprehensive agent-based system that demonstrates the application of agent planning algorithms and evaluates execution efficiency. Your design should include a detailed plan, algorithmic flow, and an evaluation report on the system's efficiency. Ensure your design contrasts with a chain-based model, highlighting the advantages of the agent-based approach.	Pass: 70 / 100	CLO-01, M10-LO1, M10-LO2, M10-LO3
<i>Formative (10.1)</i>	<i>Quiz - Chain vs Agent Models Quiz</i>	Complete a 10-question multiple-choice quiz that tests your ability to analyze and differentiate between chain-based and agent-based models. Each question presents a scenario requiring you to identify the model type and justify your choice.	<i>Pass: 80 /</i>	<i>M10-LO1</i>
<i>Formative (10.2)</i>	<i>Practice Exercise - Agent Planning Algorithm Exercise</i>	Develop a basic agent planning algorithm for a given problem scenario. Your algorithm should include a step-by-step plan and pseudocode. Focus on logical flow and applicability to the scenario.	<i>Pass: /</i>	<i>M10-LO2</i>
<i>Formative (10.3)</i>	<i>Reflection - Execution Efficiency Reflection</i>	Write a reflection on the execution efficiency of agent-based systems. Consider factors such as speed, resource usage, and scalability. Reflect on how these factors impact system performance and provide examples.	<i>Pass: /</i>	<i>M10-LO3</i>
M11 - Phase 11 - LangGraph				

Assessment	Type / Title	Task Description	Passing	Mapped LOs
Summative	Assignment - Recommendation Memo	Compose a recommendation memo evaluating the durable execution strategies in LangGraph. Your memo should include a critical analysis of current strategies, propose improvements, and justify your recommendations with evidence from course materials and external research. Address potential challenges and suggest solutions.	Pass: 70 / 100	CLO-01, M11-LO3
<i>Formative (11.1)</i>	<i>Quiz - LangGraph Architecture Quiz</i>	Complete a 10-question multiple-choice quiz focusing on the architecture of LangGraph within state machines. Questions will test your understanding of key components and their functions.	<i>Pass: 80 /</i>	<i>M11-LO1</i>
<i>Formative (11.2)</i>	<i>Practice Exercise - Agent Workflow Design</i>	Design a workflow for an agent using LangGraph. Your design should include a flowchart and a brief explanation of each step in the process. Focus on efficiency and clarity.	<i>Pass: /</i>	<i>M11-LO2</i>
<i>Formative (11.3)</i>	<i>Reflection - Durable Execution Strategies Reflection</i>	Write a reflection on the durable execution strategies in LangGraph. Discuss what you learned, challenges you encountered, and how you might apply these strategies in real-world scenarios.	<i>Pass: /</i>	<i>M11-LO3</i>
M12 - OpenAI Agents SDK				
Summative	Project - Design Artifact: Scalable AI System	Design and develop a scalable AI system using the OpenAI Agents SDK. Your project should include a detailed architecture diagram, a written explanation of your design choices, and a demonstration of the system's scalability. Ensure your design addresses potential deployment challenges and includes strategies for effective deployment.	Pass: 70 / 100	CLO-01, M12-LO1, M12-LO2, M12-LO3
<i>Formative (12.1)</i>	<i>Quiz - Architecture Analysis Quiz</i>	Complete a 10-question multiple-choice quiz focusing on the architecture of the OpenAI Agents SDK. Questions will test your understanding of key components and their interactions within the SDK.	<i>Pass: 80 /</i>	<i>M12-LO1</i>
<i>Formative (12.2)</i>	<i>Practice - Scalable System Design Exercise</i>	Design a basic scalable AI system using the OpenAI Agents SDK. Focus on creating a system that can handle increased loads and provide a brief written explanation of your design choices.	<i>Pass: /</i>	<i>M12-LO2</i>
<i>Formative (12.3)</i>	<i>Reflection - Deployment Strategy Reflection</i>	Write a reflective piece on the deployment strategies for AI applications using the OpenAI Agents SDK. Discuss potential challenges and how you would address them.	<i>Pass: /</i>	<i>M12-LO3</i>
M13 - Claude Ecosystem				
Summative	Project - Claude Ecosystem Design Artifact	Design a comprehensive artifact that demonstrates your understanding of the Claude ecosystem. This includes analyzing its architecture, creating a prompt caching strategy, and evaluating the effectiveness of extended thinking. Your artifact should integrate these elements into a cohesive design proposal that addresses potential challenges and offers innovative solutions.	Pass: 70 / 100	CLO-01, M13-LO1, M13-LO2, M13-LO3

Assessment	Type / Title	Task Description	Passing	Mapped LOs
<i>Formative (13.1)</i>	<i>Quiz - Claude Ecosystem Architecture Quiz</i>	Complete a 10-question multiple-choice quiz that tests your understanding of the Claude ecosystem architecture. Each question will assess your ability to identify and explain key components and their interactions.	<i>Pass: 80 /</i>	<i>M13-LO1</i>
<i>Formative (13.2)</i>	<i>Practice Exercise - Prompt Caching Strategy Development</i>	Develop a prompt caching strategy for the Claude ecosystem. This exercise requires you to draft a plan that outlines the steps and considerations for implementing an effective caching strategy.	<i>Pass: /</i>	<i>M13-LO2</i>
<i>Formative (13.3)</i>	<i>Reflection - Extended Thinking Effectiveness Reflection</i>	Write a reflective piece evaluating the effectiveness of extended thinking within the Claude ecosystem. Consider how extended thinking impacts decision-making and problem-solving.	<i>Pass: /</i>	<i>M13-LO3</i>
M14 - Voice AI				
Summative	Design Artifact - Voice AI System Design Project	Design a comprehensive voice AI system that demonstrates your understanding of system architecture, scalability, and deployment strategies. Your design should include a detailed architecture diagram, a plan for scalability to handle real-time interactions, and a deployment strategy that addresses potential challenges. Provide a rationale for each component of your design, supported by research and best practices in the field.	Pass: 70 / 100	CLO-01, M14-LO1, M14-LO2, M14-LO3
<i>Formative (14.1)</i>	<i>Quiz - Voice AI Architecture Quiz</i>	Complete a 10-question multiple-choice quiz to assess your understanding of voice AI system architecture. Each question will test your ability to identify and explain key components and their functions within a voice AI system.	<i>Pass: 80 /</i>	<i>M14-LO1</i>
<i>Formative (14.2)</i>	<i>Practice Exercise - Scalability Planning Exercise</i>	Develop a scalability plan for a hypothetical voice AI system designed to handle a large number of real-time interactions. Consider factors such as load balancing, data processing, and system redundancy. Submit a brief report outlining your plan and the reasoning behind your choices.	<i>Pass: /</i>	<i>M14-LO2</i>
<i>Formative (14.3)</i>	<i>Reflection - Deployment Strategy Reflection</i>	Write a reflective piece on the challenges and considerations involved in deploying voice AI systems. Discuss potential risks and how they can be mitigated. Reflect on how your understanding of deployment strategies has evolved.	<i>Pass: /</i>	<i>M14-LO3</i>
M15 - Phase 15 - Memory Systems				
Summative	Design Artifact - Memory System Design Artifact	Design a comprehensive memory system architecture that addresses scalability and performance optimization. Include a detailed schematic of the architecture, a written explanation of the design choices, and a performance evaluation report. Ensure the design is innovative and addresses potential real-world challenges.	Pass: 70 / 100	CLO-01, M15-LO1, M15-LO2, M15-LO3

Assessment	Type / Title	Task Description	Passing	Mapped LOs
<i>Formative (15.1)</i>	<i>Quiz - Memory System Architecture Quiz</i>	Complete a 10-question multiple-choice quiz focused on identifying and analyzing different memory system architectures. Each question requires selecting the correct architecture type based on given characteristics.	<i>Pass: 80 /</i>	<i>M15-LO1</i>
<i>Formative (15.2)</i>	<i>Practice Exercise - Scalable Knowledge Graph Implementation</i>	Develop a small-scale knowledge graph implementation that demonstrates scalability. Use provided datasets to create nodes and relationships, ensuring the graph can be expanded efficiently.	<i>Pass: /</i>	<i>M15-LO2</i>
<i>Formative (15.3)</i>	<i>Reflection - Memory System Performance Reflection</i>	Write a reflective piece on the performance evaluation of memory systems. Discuss the key metrics used, challenges faced during evaluation, and insights gained.	<i>Pass: /</i>	<i>M15-LO3</i>
M16 - Phase 16 - MCP				
Summative	Project - Design Artifact for MCP Implementation	Design a comprehensive artifact that demonstrates your ability to analyze MCP architecture, create a scalable client-server model, and evaluate server deployment strategies. Your artifact should include a detailed architectural diagram, a scalable model prototype, and a deployment strategy report. Ensure that each component is well-documented and justified with evidence from your research.	Pass: 70 / 100	CLO-01, M16-LO1, M16-LO2, M16-LO3
<i>Formative (16.1)</i>	<i>Quiz - MCP Architecture Analysis Quiz</i>	Complete a 10-question multiple-choice quiz assessing your understanding of MCP architecture. Each question will test your ability to identify key components and their functions within the MCP framework.	<i>Pass: 80 /</i>	<i>M16-LO1</i>
<i>Formative (16.2)</i>	<i>Practice Exercise - Scalable MCP Model Development</i>	Develop a prototype of a scalable client-server model using MCP. Your prototype should demonstrate scalability and include a brief explanation of your design choices.	<i>Pass: /</i>	<i>M16-LO2</i>
<i>Formative (16.3)</i>	<i>Reflection - Reflection on MCP Deployment Strategies</i>	Write a reflective piece on the various MCP server deployment strategies. Discuss the strengths and weaknesses of each strategy and reflect on which strategy you would recommend and why.	<i>Pass: /</i>	<i>M16-LO3</i>
M17 - AI Product Engineering				
Summative	Project - AI Product Design Artifact	Design a comprehensive AI product that demonstrates observability, cost optimization, and security measures. Your design should include a detailed plan for implementing observability in AI models, a cost optimization strategy, and an evaluation of security protocols. Provide a rationale for each component and how it aligns with industry best practices.	Pass: 70 / 100	CLO-01, M17-LO1, M17-LO2, M17-LO3
<i>Formative (17.1)</i>	<i>Quiz - Observability in AI Models Quiz</i>	Complete a 10-question multiple-choice quiz assessing your understanding of observability in AI models. Each question will test your ability to identify key components and impacts of observability on model performance.	<i>Pass: 80 /</i>	<i>M17-LO1</i>

Assessment	Type / Title	Task Description	Passing	Mapped LOs
<i>Formative (17.2)</i>	<i>Practice Exercise - Cost Optimization Plan Exercise</i>	Develop a draft cost optimization plan for a hypothetical AI system. Consider factors such as resource allocation, scaling, and budget constraints. Provide a brief rationale for your choices.	<i>Pass: /</i>	<i>M17-LO2</i>
<i>Formative (17.3)</i>	<i>Reflection - Security Measures Reflection</i>	Write a reflective piece on the importance of security measures in AI deployment. Discuss potential risks and how they can be mitigated. Reflect on how security considerations influence AI product design.	<i>Pass: /</i>	<i>M17-LO3</i>
M18 - Monetization				
Summative	Design Artifact - Monetization Strategy Design for AI SaaS Product	Design a comprehensive monetization strategy for a hypothetical AI SaaS product. Your design should include an analysis of potential monetization strategies, an evaluation of suitable payment system integrations, and a detailed credit system model. Provide a rationale for each component, supported by market research and theoretical frameworks. Present your strategy in a professional report and accompanying presentation slides.	Pass: 70 / 100	CLO-05, M18-LO1, M18-LO2, M18-LO3
<i>Formative (18.1)</i>	<i>Quiz - Monetization Strategies Quiz</i>	Complete a 10-question multiple-choice quiz on various monetization strategies for SaaS products. Each question will present a scenario requiring you to select the most appropriate monetization strategy.	<i>Pass: 80 /</i>	<i>M18-LO1</i>
<i>Formative (18.2)</i>	<i>Reflection - Payment System Integration Reflection</i>	Write a 300-word reflection on the challenges and considerations involved in integrating payment systems for AI products. Discuss potential solutions and their implications.	<i>Pass: /</i>	<i>M18-LO2</i>
<i>Formative (18.3)</i>	<i>Practice Exercise - Credit System Model Design Exercise</i>	Design a basic credit system model for an AI product. Outline the key components and explain how it would function to support monetization.	<i>Pass: /</i>	<i>M18-LO3</i>